

Psilocybin for Depressive and Anxiety Disorders in Working-Age Adults: Protocol for a Hybrid Bibliometric Analysis and Systematic Review with Meta-Analysis of Randomized Controlled Trials (2022 to 2026)

Study protocol, version 1.0

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1. Administrative information

1.1 Registration and reporting framework

This protocol follows PRISMA-P 2015 for its structure, PRISMA-S for the documentation of search strategies, and PRISMA 2020 for the reporting of the completed review. The meta-analytic arm will be registered in PROSPERO before screening begins. The bibliometric arm is not eligible for PROSPERO and is documented through this Zenodo deposit and the associated GitHub repository. Any amendment after registration will be dated, justified and versioned in both repositories.

1.2 Support and conflicts of interest

Funding sources and declarations of interest will be stated in the final manuscript. No sponsor will have a role in study selection, analysis or interpretation.

2. Introduction

2.1 Rationale

Psilocybin has moved from early-phase exploration to phase 2 and phase 3 randomized trials for depressive and anxiety disorders in a very short period. This rapid growth justifies a dual approach: a bibliometric analysis to map the structure, growth and thematic organization of the field across three citation databases, and a systematic review with meta-analysis restricted to randomized controlled trials in working-age adults with primary depressive or anxiety disorders, published between 2022 and 2026.

2.2 Research question and objectives

Research question (PICO with a study-design variant, S): In patients aged 18 to 65 years with primary depressive and/or anxiety disorders, does psilocybin treatment, compared with placebo or usual treatments, produce better clinical outcomes than the therapeutic alternatives?

- Objective 1 (bibliometric arm): to characterize scientific production on psilocybin and depression/anxiety indexed in Web of Science, Scopus and PubMed, 2022 to 2026.
- Objective 2 (meta-analytic arm): to estimate the pooled effect of psilocybin on validated depression and anxiety severity scales in randomized controlled trials, with pre-specified moderator and sensitivity analyses.

3. Methods

3.1 Eligibility criteria (PICOS)

Table 1 presents the corrected eligibility framework. Three corrections were applied with respect to the preliminary version: (a) the "Free full text" limit was removed to avoid availability bias, with full-text access managed through the UNAM institutional library and irretrievable reports documented in the PRISMA flow diagram; (b) the subheading "therapy"[Subheading] and the "treatment" free-text block were removed from the intervention strategy, since intervention relevance is more reliably established at screening than through restrictive indexing terms; (c) preprints, which are accepted at the study-design level but are poorly covered by the primary databases, are retrieved through a complementary search of Europe PMC including medRxiv, declared in Section 3.2.

Table 1. Eligibility criteria according to the PICOS framework.

Element	Inclusion	Exclusion	Search terms (MEDLINE syntax)
P (Population)	Adults aged 18 to 65 years (working age) with a primary diagnosis of a depressive disorder (e.g., major depressive disorder) and/or an anxiety disorder (e.g., generalized anxiety disorder). Participants required to maintain daily routine and occupational functioning without need for palliative care.	Depression or anxiety secondary to severe terminal illness, or explicitly reactive or secondary to somatic pathology such as cancer, Parkinson disease, dementia, or other serious conditions with behavioural implications, including alcoholism and eating disorders.	"depression"[Title/Abstract] OR "anxiety"[Title/Abstract] OR "Depression"[MeSH] OR "Anxiety"[MeSH]
I (Intervention)	Psilocybin therapy as the principal intervention for depressive and/or anxiety disorders, with or without assisted psychotherapy. Single or multiple oral doses, defined regimens.	Psilocybin combined with other psychedelics (ayahuasca, MDMA, LSD) or with methamphetamine; interventions centred on conduct disorders, alcoholism or eating disorders without separately reported depression/anxiety outcomes; recreational context.	"psilocybin"[Title/Abstract] OR "psilocybine"[Title/Abstract] OR "Psilocybine"[MeSH]. Note: the subheading "therapy"[Subheading] and the block "treatment"[Text Word] were removed from the executable strategy because they unduly restrict retrieval; intervention relevance is assessed at screening.
C (Comparison)	Placebo, treatment as usual (antidepressants), active treatment (other drugs or psychotherapies), or studies without comparator when	Survey-based comparisons without validated scales.	Comparator terms are not restricted in the search strategy; comparators are classified at extraction.

	eligible for pre-post analysis.		
O (Outcome)	Change in severity of depression and anxiety on validated scales, response and remission, functioning, quality of life, and adverse events, where clinically measurable symptom change is reported on standard instruments.	Results limited to analytical methodologies (pharmacokinetics, detection); biological or neuroimaging outcomes without clinical depressive/anxiety data; studies centred on other pathologies; studies that do not measure or report depression/anxiety outcomes or relevant adverse events.	Outcome terms are not included in the executable search strategy to preserve sensitivity; outcome eligibility is assessed at screening.
S (Study design)	Randomized controlled trials, phases 2, 3 and 4, published 2022 to 2026. Validated severity measures for depression (e.g., MADRS, HAM-D, PHQ-9, BDI) and/or anxiety (e.g., HAM-A, GAD-7, STAI, BAI). Preprints under review (retrieved through complementary preprint sources, Section 3.2).	Systematic reviews and meta-analyses (excluded from the meta-analytic arm only; eligible for the bibliometric arm), phase 1 trials in healthy participants, animal studies, short follow-up case reports, records whose full text cannot be retrieved after institutional access and author contact (documented as "reports not retrieved" in the PRISMA flow).	NOT ("systematic review"[Publication Type] OR "meta analysis"[Publication Type]). Limits: Humans, 2022 to 2026. Note: the "Free full text" limit present in a preliminary version was removed because it introduces availability bias; full-text access is managed through the UNAM institutional library.

3.2 Information sources

Bibliometric arm: Web of Science Core Collection, Scopus and PubMed. Meta-analytic arm: PubMed and Cochrane CENTRAL as primary sources, complemented by Europe PMC (which indexes medRxiv) for preprints under review. Reference lists of included trials and of recent systematic reviews will be checked manually (backward citation searching). All searches will be run on the same day and fully documented following PRISMA-S (Table 2).

Table 2. Information sources and export formats (PRISMA-S documentation).

Database / source	Interface and coverage	Role in the project	Export format
Web of Science Core Collection	Clarivate, all indexes	Bibliometric arm	Plain Text and BibTeX, full record with cited references (CR field)
Scopus	Elsevier	Bibliometric arm	CSV and BibTeX, full record
PubMed	NLM	Both arms (broad string for bibliometrics, restrictive string for the meta-analysis)	MEDLINE (.nbib)

Cochrane CENTRAL	Cochrane Library, Trials tab	Meta-analytic arm	RIS
Europe PMC (including medRxiv preprints)	europepmc.org	Complementary source of the meta-analytic arm for preprints under review	RIS / CSV

3.3 Search strategies

Bibliometric arm (broad strategies, no study-design filter; reviews are eligible for this arm):

Web of Science Core Collection (advanced search):

```
TS=(psilocybin* AND (depress* OR anxiet* OR anxious))
```

Refined by: PY=2022-2026; document types Article, Review, Early Access.

Scopus:

```
TITLE-ABS-KEY(psilocybin* AND (depress* OR anxiet* OR anxious)) AND  
PUBYEAR > 2021 AND PUBYEAR < 2027
```

PubMed (bibliometric string):

```
("Psilocybine"[MeSH] OR psilocyb*[tiab]) AND ("Depression"[MeSH] OR  
"Depressive Disorder"[MeSH] OR "Anxiety"[MeSH] OR "Anxiety  
Disorders"[MeSH] OR depress*[tiab] OR anxiet*[tiab]) AND 2022:2026[dp]
```

Meta-analytic arm (restrictive strategies):

PubMed:

```
("Psilocybine"[MeSH] OR psilocyb*[tiab]) AND ("Depression"[MeSH] OR  
"Depressive Disorder"[MeSH] OR "Anxiety"[MeSH] OR "Anxiety  
Disorders"[MeSH] OR depress*[tiab] OR anxiet*[tiab]) AND (randomized  
controlled trial[pt] OR "clinical trial, phase ii"[pt] OR "clinical trial,  
phase iii"[pt] OR "clinical trial, phase iv"[pt] OR randomi*[tiab]) NOT  
(systematic review[pt] OR meta-analysis[pt]) AND humans[mh] AND  
2022:2026[dp]
```

Limits applied: Humans; publication dates 2022 to 2026. No language limit and no full-text availability limit are applied.

Cochrane CENTRAL (advanced search, Trials tab):

```
#1 (psilocybin*):ti,ab,kw  
#2 (depress* OR anxiet* OR anxious):ti,ab,kw  
#3 #1 AND #2, publication date 2022-2026
```

Europe PMC (complementary preprint search):

```
(psilocybin*) AND (depress* OR anxiet* OR anxious) AND (SRC:PPR) AND  
(FIRST_PDATE:[2022-01-01 TO 2026-12-31])
```

The SRC:PPR filter restricts retrieval to preprint records, including medRxiv. Preprints that reach journal publication before data extraction will be replaced by the peer-reviewed version.

3.4 Study records: data management and selection

Records will be exported in the formats of Table 2. Bibliometric deduplication will follow a tri-database pipeline in R (bibliometrix), with source priority Web of Science over Scopus over PubMed based on metadata richness, reporting duplicates removed per database pair. Meta-analytic deduplication (PubMed, CENTRAL and Europe PMC) will be performed in Rayyan or Zotero.

Two reviewers will screen independently at title/abstract and full-text stages. Inter-rater agreement will be quantified with Cohen kappa; if the prevalence of inclusion is very low, Gwet AC1 will be reported alongside kappa to avoid the prevalence paradox. Disagreements will be resolved by consensus or by a third reviewer. Selection will be reported in PRISMA 2020 flow diagrams, one per arm, including the count of reports not retrieved after institutional access and author contact.

3.5 Data items and outcomes

A standardized extraction workbook will capture: study identifiers, design and phase, country, sample size, population characteristics, psilocybin dose and regimen, presence and type of assisted psychotherapy, comparator, scales used, timepoints, means and dispersion measures for depression and anxiety severity, response and remission, functioning, quality of life, and adverse events. Primary outcome: change in depression severity (MADRS, HAM-D, PHQ-9 or BDI). Secondary outcomes: change in anxiety severity (HAM-A, GAD-7, STAI, BAI), response, remission, and adverse events.

3.6 Risk of bias

Risk of bias will be assessed with RoB 2.0 by two independent reviewers at the outcome level, with results tabulated per trial and per domain. Blinding integrity deserves particular attention in psychedelic trials and will be recorded as part of domain 4 considerations and discussed narratively.

3.7 Data synthesis

Random-effects meta-analysis with REML estimation will be used. When trials report the same scale, mean differences will be pooled; across different scales, standardized mean differences (Hedges g) will be used. Heterogeneity will be described with I-squared and tau-squared, with 95 percent prediction intervals. Pre-specified moderators: dose regimen, presence of assisted psychotherapy, diagnostic population (depressive versus anxiety disorder), and comparator type. Sensitivity analyses: exclusion of high risk of bias trials and exclusion of preprints. Small-study effects will be examined with funnel plots and the Egger test when at least ten comparisons are available. Any post-hoc analysis will be explicitly declared as such.

3.8 Certainty of evidence

Certainty will be rated per outcome with GRADE, considering risk of bias, inconsistency, indirectness, imprecision and publication bias, and summarized in a Summary of Findings table.

3.9 Bibliometric arm analyses

The bibliometric corpus will be analysed in R (bibliometrix) and VOSviewer: annual production and growth rate, leading countries, institutions, sources and authors, keyword co-occurrence with thematic mapping, and citation structure using the Web of Science cited-references field. Author keywords will be cleaned with a stopword list, and network edge weights will be pruned for readability. All scripts will be released in the project repository.

4. Open science and data availability

This protocol, the search strategies, the extraction workbook template, the RoB 2.0 file, the R scripts for both arms and all derived CSV outputs will be deposited in a public GitHub repository archived in Zenodo with a citable DOI, under an MIT license for code and CC-BY for documents. Figures will be produced at 600 dpi in TIFF format for journal submission.

5. References

- [1] Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. doi:10.1136/bmj.n71
- [2] Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015;350:g7647. doi:10.1136/bmj.g7647

[3] Rethlefsen ML, Kirtley S, Waffenschmidt S, et al. PRISMA-S: an extension to the PRISMA statement for reporting literature searches in systematic reviews. *Syst Rev.* 2021;10:39. doi:10.1186/s13643-020-01542-z

[4] Sterne JAC, Savovic J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ.* 2019;366:l4898. doi:10.1136/bmj.l4898

Additional topic-specific references will be added in the final manuscript after corpus construction; in line with project standards, every reference will carry a verified DOI.